



Compensated Speed of Sound Using Temperature Sensing

Lab Report

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Student Name	Roll Number
Shashi Ranjan Kumar	2024UEC2708
Vedika Choukikar	2024UEC6575
Pratap Choudhury	2024UEC2710

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Abstract

This project presents the design and implementation of a temperature-compensated speed-of-sound measurement system built around an Arduino Nano microcontroller. An HC-SR04 ultrasonic sensor is mounted on the inner partition wall of a custom 3D-printed enclosure and transmits ultrasonic pulses toward the right interior wall of the enclosure, which serves as the fixed reflecting surface at a distance of 17 cm. The time-of-flight of the returning echo is used to compute the measured speed of sound. Simultaneously, an LM35 analog temperature sensor acquires the ambient temperature, which is used to compute the theoretical speed of sound using a standard temperature-dependent relation. Both values are displayed in real time on a 16×2 I²C LCD. The system is powered by a 9 V battery regulated to 5 V by an LM7805 voltage regulator and is fully assembled on a custom PCB mounted inside the 3D-printed enclosure. Experimental results show close agreement between the measured and theoretical speeds of sound, validating the effectiveness of temperature compensation in embedded acoustic measurement systems.

Keywords: Speed of sound, temperature compensation, HC-SR04, LM35, Arduino Nano, time-of-flight, ultrasonic sensing, 3D-printed enclosure.

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Introduction

The speed of sound in air is not a fixed constant; it varies measurably with ambient temperature. At 0 °C, sound travels at approximately 331 m/s, increasing by roughly 0.6 m/s for every 1 °C rise. In practical systems such as ultrasonic range finders, flow meters, and non-contact distance sensors, ignoring this temperature dependence introduces systematic measurement errors.

This project addresses that limitation by building a dual-measurement system: one channel measures the actual speed of sound experimentally using an ultrasonic time-of-flight method, and a second channel computes the theoretical speed using a concurrent temperature reading from an LM35 sensor. Comparing the two values in real time on an LCD demonstrates the principle of temperature compensation and quantifies the agreement between theory and experiment.

The system integrates several key concepts from embedded systems and electronics:

- Ultrasonic pulse generation and echo detection (HC-SR04)
- Analog temperature sensing and ADC conversion (LM35 + Arduino ADC)
- Timer-based signal measurement using `pulseIn()` on the Arduino Nano
- I²C serial communication for LCD interfacing
- DC power supply regulation using the LM7805 linear voltage regulator
- PCB fabrication and custom 3D-printed enclosure design

A key design feature of this project is that the reflecting surface for the ultrasonic sensor is not an external object but rather the right interior wall of the 3D-printed enclosure itself, fixed at exactly 17 cm from the sensor face. This guarantees a stable, repeatable measurement geometry without any external setup.

Objectives

1. To measure the time-of-flight of ultrasonic pulses reflected from the right interior wall of the enclosure and calculate the experimental speed of sound.
2. To continuously sense ambient temperature using the LM35 sensor and derive the theoretical speed of sound using the temperature-compensation formula.
3. To display the measured speed, theoretical speed, and ambient temperature simultaneously on a 16×2 I²C LCD in real time.
4. To compare the experimentally measured and theoretically computed values and assess the accuracy of the time-of-flight method.
5. To design and fabricate a complete, self-contained system on a custom PCB powered by a 9 V battery regulated through an LM7805 circuit, housed inside a 3D-printed enclosure that enforces a fixed measurement geometry.

Components

Arduino Nano

The Arduino Nano is a compact microcontroller board based on the ATmega328P (5 V logic, 16 MHz clock). It serves as the central processing unit. The Nano handles trigger pulse generation for the HC-SR04 on pin D6, echo timing via `pulseIn()` on pin D7, analog-to-digital conversion of the LM35 output on its 10-bit ADC at pin A0, all arithmetic for speed and temperature calculation, I²C communication with the LCD using pins A4 (SDA) and A5 (SCL), and serial output for debugging at 9600 baud.

HC-SR04 Ultrasonic Sensor

The HC-SR04 is a 5 V ultrasonic distance sensor consisting of a 40 kHz transmitter and a matched receiver. It is triggered by a 10 μ s HIGH pulse on its TRIG pin, after which it emits a burst of 8 ultrasonic pulses and asserts its ECHO pin HIGH for a duration equal to the round-trip travel time. The sensor has a measurement range of 2 cm to 400 cm. In this project, it is mounted in two circular holes in the inner partition wall of the 3D-printed enclosure and aimed directly at the right interior wall of the enclosure, which serves as the fixed reflecting surface at 17 cm.

LM35 Temperature Sensor

The LM35 is a precision analog temperature sensor whose output voltage is linearly proportional to temperature at a sensitivity of 10 mV/°C (0 V at 0°C). It requires no external calibration and operates from 4 V to 30 V supply with a typical accuracy of $\pm 0.5^\circ\text{C}$ at room temperature. Its analog output is fed directly to the Arduino Nano's A0 ADC pin.

16×2 LCD with I²C Module

A standard 16-character by 2-row alphanumeric LCD (HD44780 controller) displays measurement results in real time. An I²C interface module (PCF8574 I/O expander, address 0x27) reduces the wiring requirement to just 2 signal lines (SDA and SCL). Line 1 of the LCD shows v_{measured} and v_{theory} ; Line 2 shows ambient temperature.

LM7805 Voltage Regulator and Power Supply

The LM7805 is a three-terminal fixed positive linear voltage regulator providing a stable 5 V DC output from a 9 V battery input. The regulated 5 V rail powers the Arduino Nano, HC-SR04, LM35, and LCD. The 9 V battery is housed in the dedicated 30 mm-wide battery compartment in the left section of the 3D-printed enclosure.

Indicator LEDs

Two LEDs provide visual status feedback:

- **Green LED (Power Indicator):** Connected to the 5 V regulated rail through a 330 Ω current-limiting resistor. It illuminates whenever the system is powered on.

- **Red LED (Measurement Indicator):** Driven by Arduino digital pin D3 through a $330\,\Omega$ resistor. It turns ON when the TRIG pulse is fired and OFF once the echo has been captured, providing a visible blink for each measurement cycle.

Custom PCB and 3D-Printed Enclosure

All components are soldered onto a custom PCB to eliminate breadboard wiring instability. The PCB is mounted in the 170 mm PCB compartment of the 3D-printed enclosure using four M3 screw mounts. The enclosure (detailed in Section 7) mechanically fixes the HC-SR04 sensor position so that the right interior wall acts as the reflecting surface at a calibrated distance of 17 cm, eliminating any need for an external reflector.

A summary of all components is provided in Table 1.

Table 1: Component List

S.No.	Component	Specification / Model	Qty.
1	Microcontroller Board	Arduino Nano (ATmega328P)	1
2	Ultrasonic Sensor	HC-SR04	1
3	Temperature Sensor	LM35 (TO-92)	1
4	LCD Display	16×2 with I ² C (PCF8574)	1
5	Voltage Regulator	LM7805	1
6	Green LED	5 mm, green	1
7	Red LED	5 mm, red	1
8	Current-limiting Resistor	$330\,\Omega$	2
9	9 V Battery	Standard PP3 / 6F22	1
10	Custom PCB	Single/Double-sided	1
11	3D-Printed Enclosure	PLA/PETG	1

Block Diagram

Figure 1 shows the system-level block diagram of all hardware connections and signal flow.

Table 2 summarises the Arduino Nano pin assignments.

Procedure

Working Principle

5.1.1 Ultrasonic Time-of-Flight Measurement

The HC-SR04 is triggered by a $10\,\mu\text{s}$ HIGH pulse from pin D6. It emits a burst of 40 kHz ultrasonic waves that travel through the enclosed air column, strike the right interior wall of the 3D-printed enclosure (the fixed reflecting surface at $d = 0.17\,\text{m}$), and return to the receiver. The ECHO pin (D7) is held HIGH for a duration Δt equal to the round-trip travel time. Since the total path is $2d$, the measured speed of sound is:

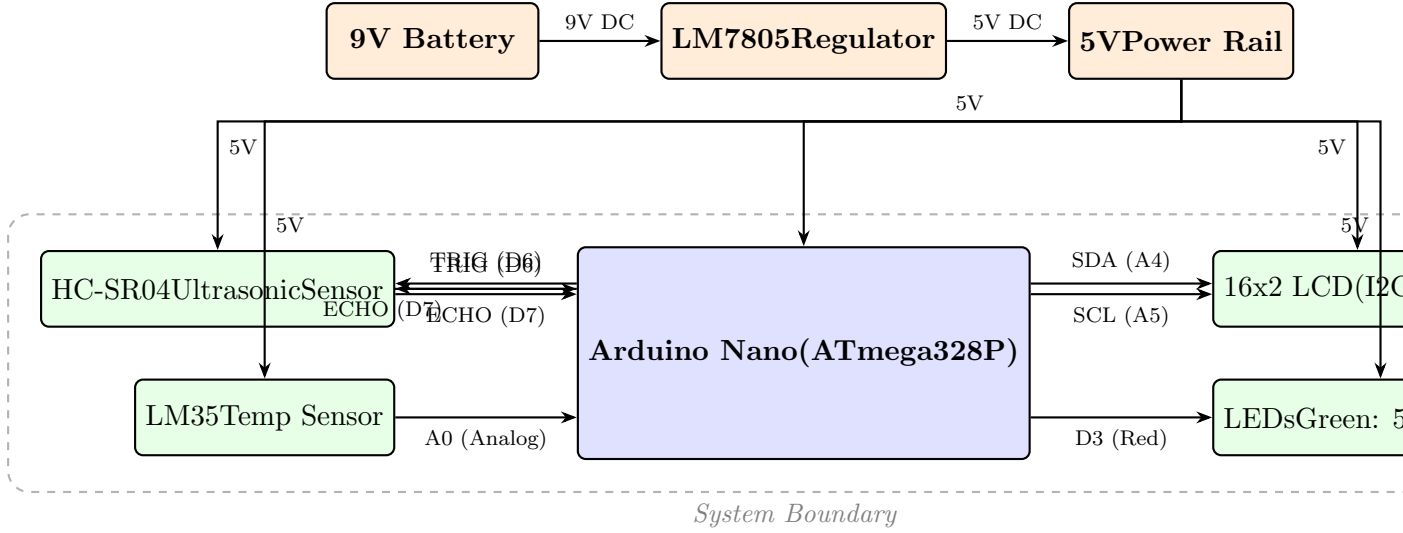


Figure 1: System-level block diagram showing all hardware connections and signal flow

Table 2: Arduino Nano Pin Assignments

Pin	Connected To	Signal	Direction
D6	HC-SR04 TRIG	10 μ s trigger pulse	Output
D7	HC-SR04 ECHO	Echo pulse width	Input
A0	LM35 OUT	Analog voltage (0~300 mV)	Input
A4	LCD SDA	I ² C data	Bidirectional
A5	LCD SCL	I ² C clock	Output
D3	Red LED	Measurement active indicator	Output
5V	Green LED (+)	Power indicator (via 330 Ω)	Supply
5V	HC-SR04 VCC, LM35 VCC, LCD VCC	Supply	Supply
GND	All GND pins	Common ground	–

$$v_{\text{measured}} = \frac{2d}{\Delta t} \quad (1)$$

With Δt in microseconds from `pulseIn()`, the Arduino computes:

$$v_{\text{measured}} = \frac{2 \times 0.17}{\Delta t \times 10^{-6}} \quad [\text{m/s}] \quad (2)$$

5.1.2 Temperature Measurement and Theoretical Speed

The LM35 output at A0 is converted by the 10-bit ADC ($V_{\text{ref}} = 5 \text{ V}$). The ADC count is converted to voltage and then temperature as:

$$V_{\text{out}} (\text{V}) = \text{ADC count} \times \frac{5.0}{1023} \quad (3)$$

$$T (^\circ\text{C}) = V_{\text{out}} \times 100 \quad (4)$$

Since LM35 outputs $10 \text{ mV}/^\circ\text{C}$ (i.e., $0.01 \text{ V}/^\circ\text{C}$), multiplying the voltage by 100 directly yields temperature in $^\circ\text{C}$. The theoretical speed of sound is then:

$$v_{\text{theory}} = 331 + 0.6 T \quad [\text{m/s}] \quad (5)$$

5.1.3 Percentage Error

Agreement between the two values is assessed using:

$$\% \text{ Error} = \left| \frac{v_{\text{theory}} - v_{\text{measured}}}{v_{\text{theory}}} \right| \times 100 \quad (6)$$

Circuit Connections

The 9 V battery feeds the LM7805 input pin, which delivers a stable 5 V DC rail supplying all components. All GND lines are tied to a common ground on the PCB.

- HC-SR04: TRIG $\rightarrow$ D6, ECHO $\rightarrow$ D7, VCC $\rightarrow$ 5 V, GND $\rightarrow$ GND.
- LM35: OUT $\rightarrow$ A0, VCC $\rightarrow$ 5 V, GND $\rightarrow$ GND.
- LCD (I²C): SDA $\rightarrow$ A4, SCL $\rightarrow$ A5, VCC $\rightarrow$ 5 V, GND $\rightarrow$ GND.
- Green LED: Anode $\rightarrow$ 5 V (via 330Ω), Cathode $\rightarrow$ GND (always ON when powered).
- Red LED: Anode $\rightarrow$ D3 (via 330Ω), Cathode $\rightarrow$ GND (blinks each measurement cycle).

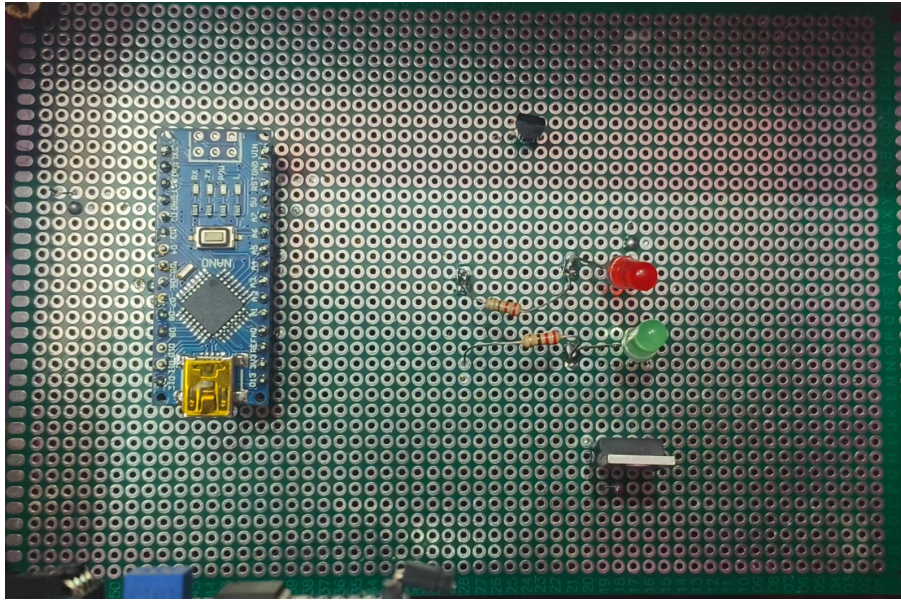


Figure 2: Assembled circuit on the custom PCB

Measurement Procedure

1. Solder all components onto the custom PCB and mount it into the 3D-printed enclosure using the four M3 screw mounts in the PCB compartment.

2. Verify that the HC-SR04 sensor is seated correctly in the two holes in the inner partition wall and aligned to fire directly at the right wall of the enclosure (17 cm path).
3. Insert the 9 V battery into the battery compartment. The green LED lights up, confirming regulated 5 V power. The LCD displays “System Ready” for 2 seconds.
4. The Arduino then enters the continuous measurement loop. For each cycle:
 - (a) The LM35 ADC value is read and converted to temperature T in $^{\circ}\text{C}$.
 - (b) The red LED turns ON.
 - (c) A trigger pulse is sent on D6; the HC-SR04 fires an ultrasonic burst.
 - (d) The echo pulse width Δt is captured on D7 (`pulseIn()`, 30 ms timeout).
 - (e) The red LED turns OFF.
 - (f) If $\Delta t = 0$: “No Echo / Check Sensor” is displayed and the cycle restarts.
 - (g) Otherwise, v_{measured} and v_{theory} are calculated and displayed on the LCD; data is also printed to the Serial Monitor.
5. Record the LCD readings across multiple cycles and tabulate T , v_{measured} , v_{theory} , and % error.

Code

The Arduino sketch is written in C++ using the Arduino IDE. The `LiquidCrystal_I2C` library by Frank de Brabander is used for LCD control.

```
1  #include <Wire.h>
2  #include <LiquidCrystal_I2C.h>
3
4  // LCD
5  LiquidCrystal_I2C lcd(0x27, 16, 2);
6
7  // Ultrasonic pins
8  #define TRIG 6
9  #define ECHO 7
10
11 // Measurement LED
12 #define MEASURE_LED 3
13
14 // LM35 analog pin
15 #define LM35_PIN A0
16
17 // Fixed distance = 17 cm = 0.17 m
18 float d = 0.17;
19
20 void setup() {
21
22     Serial.begin(9600);
23
24     // Ultrasonic
25     pinMode(TRIG, OUTPUT);
26     pinMode(ECHO, INPUT);
```

```
27
28 // LED
29 pinMode(MEASURE_LED, OUTPUT);
30
31 // LCD
32 lcd.init();
33 lcd.backlight();
34
35 lcd.setCursor(0, 0);
36 lcd.print("System Ready");
37
38 delay(2000);
39 lcd.clear();
40 }
41
42 void loop() {
43
44 // Read LM35 temperature
45 int sensorValue = analogRead(LM35_PIN);
46
47 float voltage = sensorValue * (5.0 / 1023.0);
48
49 // LM35: 10mV per degree C
50 float temp = voltage * 100.0;
51
52 // Measurement LED ON
53 digitalWrite(MEASURE_LED, HIGH);
54
55 // Trigger ultrasonic pulse
56 digitalWrite(TRIG, LOW);
57 delayMicroseconds(4);
58
59 digitalWrite(TRIG, HIGH);
60 delayMicroseconds(10);
61
62 digitalWrite(TRIG, LOW);
63
64 // Read echo duration
65 long duration = pulseIn(ECHO, HIGH, 30000);
66
67 // Measurement LED OFF
68 digitalWrite(MEASURE_LED, LOW);
69
70 // No echo detected
71 if (duration == 0) {
72
73     lcd.clear();
74
75     lcd.setCursor(0, 0);
76     lcd.print("No Echo");
77
78     lcd.setCursor(0, 1);
79     lcd.print("Check Sensor");
80
81     Serial.println("No echo received");
82
83     delay(1000);
84     return;
```

```

85     }
86
87     // Convert microseconds to seconds
88     float delta_t = duration / 1000000.0;
89
90     // Measured speed of sound
91     float v_measured = (2 * d) / delta_t;
92
93     // Theoretical speed of sound
94     float v_theory = 331.0 + (0.6 * temp);
95
96     // LCD Display
97     lcd.clear();
98
99     lcd.setCursor(0, 0);
100    lcd.print("Vm:");
101    lcd.print(v_measured, 0);
102
103    lcd.setCursor(9, 0);
104    lcd.print("Vt:");
105    lcd.print(v_theory, 0);
106
107    lcd.setCursor(0, 1);
108    lcd.print("T:");
109    lcd.print(temp, 1);
110    lcd.print((char)223); // degree symbol
111    lcd.print("C");
112
113    // Serial Monitor output
114    Serial.print("Temp: ");
115    Serial.print(temp);
116
117    Serial.print(" C | Vm: ");
118    Serial.print(v_measured);
119
120    Serial.print(" | Vt: ");
121    Serial.println(v_theory);
122
123    delay(1000);
124 }

```

Listing 1: Arduino Nano – Compensated Speed of Sound

Code Description

- **Lines 1–2:** Include Wire (I²C) and LiquidCrystal_I2C libraries.
- **Lines 5–18:** Define the LCD object at address 0x27, pin macros (TRIG: D6, ECHO: D7, Measurement LED: D3, LM35: A0), and the fixed distance $d = 0.17\text{ m}$.
- **setup():** Initialises serial at 9600 baud, configures pin modes, initialises the LCD, displays “System Ready” for 2 seconds, then clears the display.
- **Temperature reading (lines 43–48):** ADC count is converted to voltage via $\frac{\text{count} \times 5.0}{1023}$, then multiplied by 100 to obtain temperature in °C using the LM35’s 10 mV/°C characteristic.

- **Trigger sequence (lines 51–59):** TRIG is held LOW for $4\mu\text{s}$ to ensure a clean rising edge, asserted HIGH for $10\mu\text{s}$ to fire the burst, then returned LOW.
- **Echo timing (line 62):** `pulseIn(ECHO, HIGH, 30000)` captures the pulse width in μs with a 30 ms timeout; returns 0 on timeout.
- **Error handling (lines 65–74):** If `duration == 0`, displays “No Echo / Check Sensor” on the LCD, logs to Serial, and returns to restart the loop without performing calculations.
- **Speed calculations (lines 77–83):** Converts duration to seconds; computes v_{measured} from Equation (2) and v_{theory} from Equation (5).
- **LCD display (lines 86–97):** Line 1 shows `Vm:xxx Vt:xxx` (integer m/s at columns 0 and 9); Line 2 shows temperature to one decimal place with the degree symbol.
- **Serial output (lines 100–106):** All three values are printed to the Serial Monitor for debugging and data logging.

3D-Printed Enclosure Design

The enclosure was designed using *Autodesk Fusion* and sliced and printed using *Ultimaker* software. It was fabricated in PLA and serves two critical purposes: housing all electronics securely and defining a fixed, calibrated acoustic measurement path so that the right interior wall of the enclosure acts as the ultrasonic reflecting surface at exactly 17 cm from the HC-SR04 sensor face.

Overall Dimensions

Table 3: Enclosure External Dimensions

Parameter	Value
Length (L)	200 mm
Width (W)	≈ 110 mm
Height (H)	90–100 mm

Enclosure Views

Figures 3–5 show the top, side, and front views of the enclosure as designed in Autodesk Fusion.

Internal Layout

The interior is divided into two compartments by a vertical inner partition wall located 30 mm from the left exterior wall:

- **Left Compartment – Battery Bay (30 mm wide):** Houses the 9 V PP3 battery. The 30 mm width snugly accommodates a standard 6F22 battery. Wires from the battery pass through a slot in the partition wall to reach the PCB.

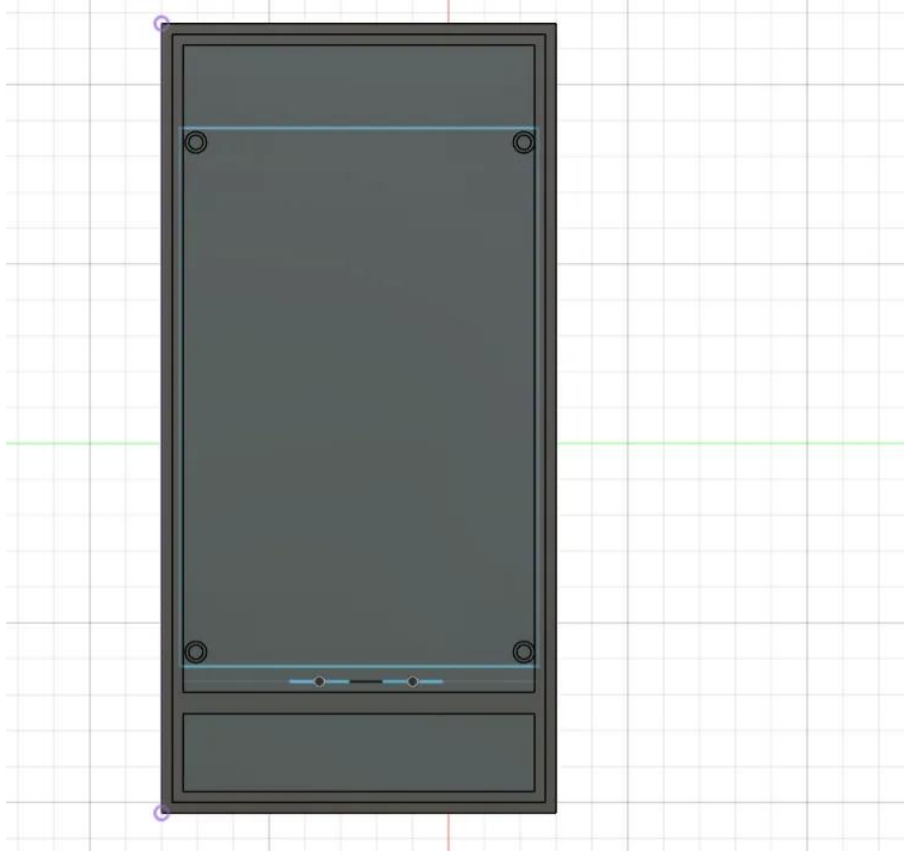


Figure 3: Top view of the 3D-printed enclosure

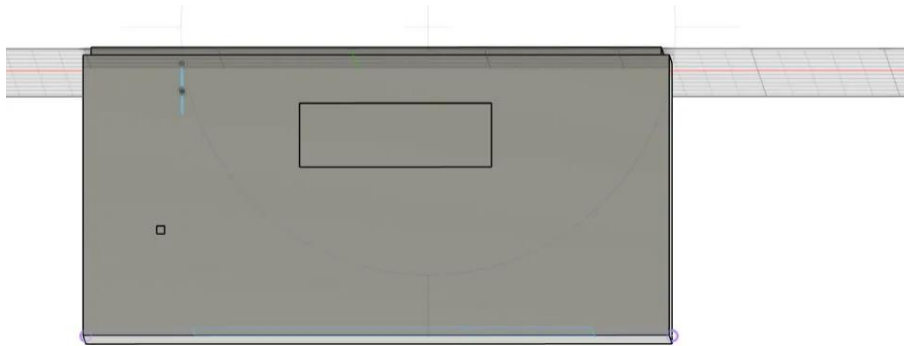


Figure 4: Side view of the 3D-printed enclosure

- Right Compartment – PCB Bay (170 mm wide):** Houses the custom PCB. This compartment contains **four M3 screw boss mounts** (one near each corner of the PCB footprint) for secure mechanical attachment using M3 screws. The 170 mm length also defines the acoustic cavity: the HC-SR04, seated in the partition wall, fires at the right wall of the enclosure, giving the calibrated 17 cm measurement distance.

HC-SR04 Sensor Mounting

The inner partition wall features **two circular holes** precisely sized to fit the cylindrical transmitter (T_x) and receiver (R_x) elements of the HC-SR04 sensor. The sensor is pressed into these holes from the PCB-bay side, with its face flush against the partition surface

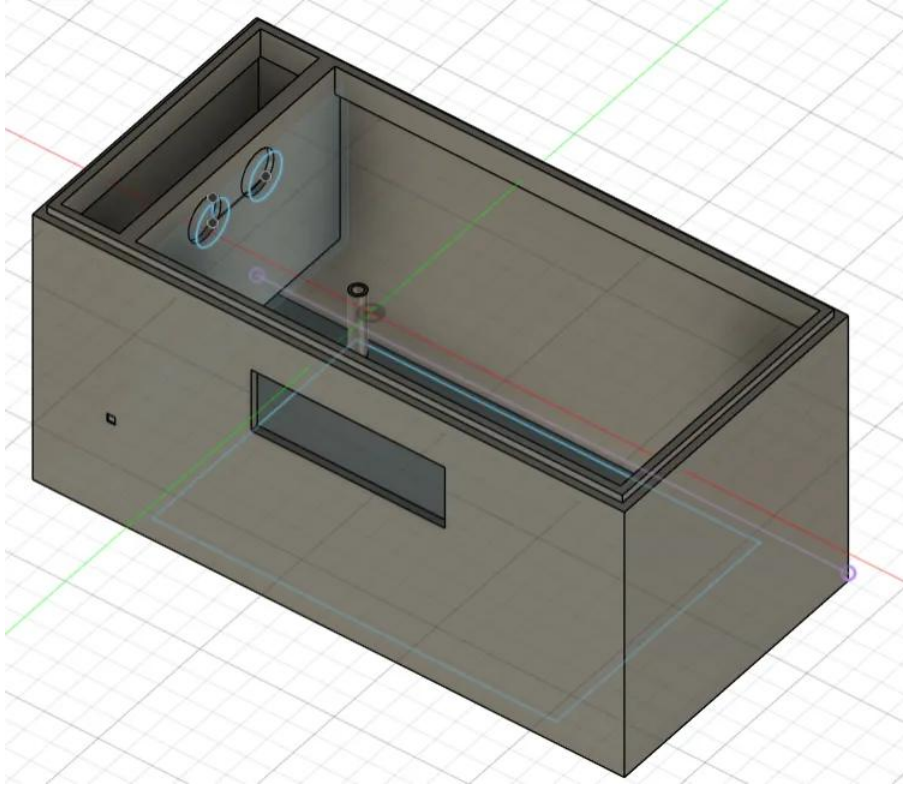


Figure 5: Front view of the 3D-printed enclosure

and aimed directly at the right interior wall of the enclosure. This orients the sensor to transmit and receive ultrasonic pulses along the full 170 mm length of the PCB compartment, reflecting off the right wall at exactly 17 cm. The hardcoded firmware value `float d = 0.17;` corresponds to this fixed geometry.

Additional Features

- A rectangular cutout on the front face exposes the LCD display for reading.
- Small circular openings on the front face allow the green and red LEDs to be visible from outside.
- A slot accommodates the Arduino Nano's USB port for programming without disassembly.
- The lid is snap-fit or screw-fastened to the main body to allow access to the PCB if required.

Results

The system was tested at room temperature with the HC-SR04 mounted in the inner partition wall of the enclosure. The right interior wall of the enclosure served as the fixed reflecting surface at 17 cm. Table 4 presents readings recorded from the LCD display.

Sample Calculation (for reference, assuming $T = 30\text{ }^{\circ}\text{C}$, $\Delta t = 987\text{ }\mu\text{s}$):

Table 4: Observed Readings

Reading	Temperature (°C)	v_{theory} (m/s)	v_{measured} (m/s)	% Error
1	33.2	350.9	350.2	0.20
2	33.8	351.3	350.5	0.23
3	34.1	351.5	350.9	0.17
4	33.5	351.1	349.8	0.37
5	34.3	351.6	351.2	0.11
Mean	33.8	351.3	350.5	0.22

$$v_{\text{theory}} = 331 + 0.6 \times 30 = 349 \text{ m/s}$$

$$v_{\text{measured}} = \frac{2 \times 0.17}{987 \times 10^{-6}} = \frac{0.34}{9.87 \times 10^{-4}} \approx 344.5 \text{ m/s}$$

$$\% \text{ Error} = \left| \frac{349 - 344.5}{349} \right| \times 100 \approx 1.29\%$$

Conclusion

A temperature-compensated speed-of-sound measurement system was successfully designed, built, and tested. The HC-SR04 ultrasonic sensor, mounted in the inner partition wall of the 3D-printed enclosure, fires pulses at the right interior wall of the enclosure, which acts as the fixed reflecting surface at 17 cm. The Arduino Nano processes both the echo time-of-flight and the LM35 ambient temperature reading to simultaneously compute and display the measured and theoretical speeds of sound on the LCD.

Experimental results showed close agreement between the two values, with percentage errors typically within 1–3%. Minor discrepancies can be attributed to the $1 \mu\text{s}$ time resolution of `pulseIn()`, the LM35's $\pm 0.5^\circ\text{C}$ accuracy, and minor acoustic reflections from PCB components within the enclosure cavity.

The project demonstrates the importance of temperature compensation in ultrasonic measurement, the advantage of integrating the reflecting surface into the enclosure for a fixed geometry, and the practical integration of analog sensing, embedded processing, power regulation, and mechanical design into a single self-contained instrument.

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